



Setting Up AWS Resources: EC2, S3, and VPC

Introduction

What are AWS Resources?

AWS (Amazon Web Services) provides various **cloud computing services**.

Three of the most fundamental services are:

- **EC2 (Elastic Compute Cloud)** – Virtual servers in the cloud.
- **S3 (Simple Storage Service)** – Scalable cloud storage.
- **VPC (Virtual Private Cloud)** – A private, isolated network in AWS.

Why Use These AWS Resources?

- **EC2:** Hosts applications and services on the cloud.
- **S3:** Stores files, backups, and logs securely.
- **VPC:** Provides network security and custom IP addressing.

1. Setting Up an EC2 Instance

Step 1: Login to AWS Console

- Go to [AWS Console](#).
- Navigate to **EC2** under **Compute**.

Step 2: Launch a New EC2 Instance

1. Click **Launch Instance**.
2. Choose an **Amazon Machine Image (AMI)** (e.g., Ubuntu, Amazon Linux).
3. Select an **Instance Type** (e.g., t2.micro for free tier).
4. Configure instance settings (default is fine for now).
5. Add **Storage** (default: 8GB, can be increased).



6. Add **Tags** (optional, e.g., Name: MyEC2Instance).
7. Configure **Security Group** (Allow SSH on **port 22**).
8. Click **Launch**, then **Create a New Key Pair** and **Download it**.

Step 3: Connect to the EC2 Instance

Use SSH to connect (for Linux/macOS):

```
ssh -i my-key.pem ec2-user@<EC2-Public-IP>
```

For Windows, use **PuTTY** and load the **.pem** file.

2. Setting Up an S3 Bucket

Step 1: Create an S3 Bucket

1. Navigate to **S3** in AWS Console.
2. Click **Create Bucket**.
3. Provide a unique **Bucket Name** (e.g., **my-storage-bucket**).
4. Choose a **Region** (e.g., **us-east-1**).
5. Disable **Block Public Access** (if needed).
6. Click **Create Bucket**.

Step 2: Upload and Access Files

- Click the bucket name.
- Click **Upload** → Select a file → Click **Upload**.

Step 3: Enable Public Access (Optional)

- Go to **Permissions** → **Bucket Policy** → Add this JSON policy:

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Principal": "*",  
      "Action": "s3:*",  
      "Resource": "arn:aws:s3:::my-storage-bucket/*"  
    }  
  ]  
}
```



```
{
  "Effect": "Allow",
  "Principal": "*",
  "Action": "s3:GetObject",
  "Resource": "arn:aws:s3:::my-storage-bucket/*"
}
]
```

- Now, anyone can access the files via

<https://my-storage-bucket.s3.amazonaws.com/myfile.txt>.

3. Setting Up a VPC

Step 1: Create a VPC

1. Go to **VPC Dashboard**.
2. Click **Create VPC** → Select **VPC only**.
3. Enter **CIDR Block** (e.g., [10.0.0.0/16](#)).
4. Click **Create VPC**.

Step 2: Create Subnets

1. Navigate to **Subnets** → **Create Subnet**.
2. Select **VPC** → Choose **CIDR Block** (e.g., [10.0.1.0/24](#)).
3. Repeat for a second subnet.

Step 3: Create an Internet Gateway

1. Go to **Internet Gateways** → **Create Internet Gateway**.
2. Attach it to your VPC.



Step 4: Configure Route Table

1. Navigate to **Route Tables** → **Edit Routes**.
2. Add a rule: **Destination:** 0.0.0.0/0, **Target:** Internet Gateway.

Know More: FAQs

1. What is the difference between EC2 and S3?

Feature	EC2	S3
Purpose	Compute (Virtual Machines)	Storage (Files, Objects)
Pricing	Pay per compute (per second/minute)	Pay per GB stored
Use Case	Hosting applications, databases	Storing backups, logs, static assets

2. How do I secure my EC2 instance?

- Use **SSH key pairs** for login.
- Restrict **security group rules** to allow only necessary access.
- Enable **IAM roles** for permissions.

3. Can I access S3 without the internet?

- Yes, using **AWS PrivateLink** or **VPC Endpoints**.

4. How do I delete resources to avoid charges?

- **EC2:** Terminate instance.
- **S3:** Delete bucket.
- **VPC:** Remove subnets, internet gateway, and VPC.



Conclusion

Setting up **EC2, S3, and VPC** is the foundation of AWS cloud computing. With these resources, you can host applications, store data, and secure your network in the cloud efficiently!